

A Brief Introduction to Morse Theory

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1 Abstract

This is an expository paper on Morse theory on smooth manifolds. We begin by introducing the basic definitions needed for Morse theory and state the Morse lemma together with the two Morse theorems. Then we are focusing on explain how these theorems been reflected in differential and algebraic topology through the connection between Morse theory and CW complexes. Finally, we present several examples illustrating the applications of the theorems discussed above. The main reference is [5].

2 Basic Definition

We start with some definitions. Let M be a smooth manifold, and $f : M \rightarrow \mathbb{R}$ be a smooth function. For $a \in \mathbb{R}$, we denote the sub-level set as $M^a = f^{-1}((-\infty, a])$.

Definition 2.1 (Non-degeneracy). Let $p \in M$ be a critical point of a smooth function f on M . p is *non-degenerate* if, under the local coordinates system $(x^1, \dots, x^n)$ on some neighbourhood $U \subseteq M$ of p , the Hessian matrix

$$Hf(p) = \left(\frac{\partial^2 f}{\partial x^i \partial x^j}(p) \right)_{i,j=1}^n$$

is non-singular.

Definition 2.2 (Hessian as a bilinear form). For $p \in M$ and smooth function f on M , we define the Hessian of f at p as the bilinear form

$$Hf(p) : T_p M \times T_p M \rightarrow \mathbb{R}, Hf(p)(v, w) = V(W(f))(p)$$

where V, W are extensions of v, w as smooth vector fields on some neighbourhood of p .

If p is a critical point of f , we have

$$[V, W]_p(f) = V(W(f))(p) - W(V(f))(p) = 0,$$

i.e. the Hessian is a symmetric bilinear form. It is also well defined, i.e. independent of the choice of extensions V, W of v, w (note that we have $V(W(f))(p) = d(W(f))_p(v)$ and similarly $W(V(f))(p) = d(V(f))_p(w)$).

Definition 2.3 (Index and nullity). Let $p \in M$ be a critical point of a smooth function f . The *index* of f at p , denoted as $\text{index}_p f$, is the maximal dimension of a subspace of $T_p M$ on which $Hf(p)$ is negative definite. The *nullity* is the dimension of the subspace V such that for every $v \in V$ and $w \in T_p M$ we have $Hf(p)(v, w) = 0$.

Notice that, by definition p is non-degenerate if and only if the nullity of $Hf(p)$ is 0.

Definition 2.4 (Morse function). Let f be a smooth function on M . It is a *Morse function* if all its critical points are non-degenerate. We also say f is *strictly non-degenerate* if in addition it takes distinct value at each critical point.

As an example, in the case of a torus $T \subset \mathbb{R}^3$, a convenient choice of a Morse function is the height function. There are 4 critical points, of indices 0,1,1,2, respectively. One can see that the critical point with index 0 is the minimum point, those with index 1 saddle points, and the one with index 2 the maximum point.

3 Morse Theory for Smooth Functions

This section focuses on the local behaviour of a smooth function near non-degenerate critical points, which forms the core of Morse theory.

Lemma 3.1 (Morse's Lemma). *Let f be a smooth function on M , and $p \in M$ be a non-degenerate critical point of f of index λ . Then there exists a neighborhood $U \subset M$ of p with local coordinates $(y^1, \dots, y^n)$, such that $y^i(p) = 0$ for all $i = 1, \dots, n$ and the following holds on U :*

$$f = f(p) - \sum_{i=1}^{\lambda} (y^i)^2 + \sum_{i=\lambda+1}^n (y^i)^2.$$

For the proof, we first show the following lemma:

Lemma 3.2. *Let f is a smooth function, defined in a convex neighborhood V of $0 \in \mathbb{R}^n$. Without loss of generality suppose $f(0) = 0$. Then there exists a smooth function $g_i : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $g_i(0) = \frac{\partial f}{\partial x_i}(0)$.*

Proof. For each $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, let $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ be the curve defined by $\gamma(t) = (tx_1, \dots, tx_n)$. Then we have:

$$\begin{aligned} f(x_1, \dots, x_n) &= \int_0^1 \frac{df(tx_1, \dots, tx_n)}{dt} dt \\ &= \int_0^1 \frac{d}{dt} f \circ \gamma(t) dt \\ &= \int_0^1 \frac{\partial}{\partial x_i} (f(\gamma(t))) \cdot \frac{d}{dt} \gamma(t) dt \\ &= \int_0^1 \sum_{i=1}^n x_i \frac{\partial f}{\partial x_i} (tx_1, \dots, tx_n) dt \\ &= \sum_{i=1}^n x_i \int_0^1 \frac{\partial f}{\partial x_i} (tx_1, \dots, tx_n) dt \end{aligned}$$

Letting $g_i = \int_0^1 \frac{\partial f}{\partial x_i} (tx_1, \dots, tx_n) dt$, we obtain $f = \sum_{i=1}^n x_i g_i$ as claimed. $\square$

Proof of Lemma 3.1. As the statement is local, it suffices to assume $M = \mathbb{R}^n$ and $p = 0$. WLOG we also assume $f(p) = f(0) = 0$. By Lemma 3.2, there exist $g_j : \mathbb{R}^n \rightarrow \mathbb{R}^n, 1 \leq j \leq n$, such that

$$f(x_1, \dots, x_n) = \sum_{j=1}^n x_j g_j(x_1, \dots, x_n).$$

Since p is a critical point of f , we have:

$$g_j(0) = \frac{\partial f}{\partial x^j}(0) = 0.$$

By Lemma 3.2 again, we have:

$$g_j(x_1, \dots, x_n) = \sum_{i=1}^n x_i b_{ij}(x_1, \dots, x_n)$$

for some smooth function b_{ij} on $\mathbb{R}^n$. Hence

$$f(x_1, \dots, x_n) = \sum_{j=1}^n \sum_{i=1}^n x_i x_j b_{ij}(x_1, \dots, x_n)$$

We may replace b_{ij} with $\frac{1}{2}(b_{ij} + b_{ji})$ if necessary, and assume $b_{ij} = b_{ji}$. By construction we have $b_{ij}(0) = \frac{1}{2} \frac{\partial^2 f}{\partial x_i \partial x_j}(0)$. Since p is non-degenerate by assumption, the matrix $\left(\frac{1}{2} \frac{\partial^2 f}{\partial x_i \partial x_j}(0) \right)_{i,j}$ is non-singular, and hence there exists some $\frac{\partial^2 f}{\partial x_{i_0} \partial x_{j_0}}(0) \neq 0$. We have two cases:

Case 1: $i_0 = j_0$. WLOG we can exchange x_{i_0} with x_1 , so that $\frac{\partial^2 f}{\partial x_1 \partial x_1} = 2a_{11} \neq 0$.

Case 2: $i_0 \neq j_0$. We define new coordinates as follows:

$$\tilde{x}_k = \begin{cases} \frac{(x_{i_0} + x_{j_0})}{2} & \text{if } k = i_0 \\ \frac{(x_{i_0} - x_{j_0})}{2} & \text{if } k = j_0 \\ x_k & \text{otherwise} \end{cases}.$$

This is a linear change of coordinates and invertible, so $\tilde{x}_1, \dots, \tilde{x}_n$ also form a coordinate system in $\mathbb{R}^n$. We have

$$x_{i_0} x_{j_0} a_{i_0 j_0}(x) = (\tilde{x}_{i_0} + \tilde{x}_{j_0})(\tilde{x}_{i_0} - \tilde{x}_{j_0}) a_{i_0 j_0}(\tilde{x}_1, \dots, \tilde{x}_{i_0} \pm \tilde{x}_{j_0}, \dots, \tilde{x}_{i_0} \mp \tilde{x}_{j_0}, \dots, \tilde{x}_n)$$

i.e. $\frac{\partial^2 f}{\partial \tilde{x}_{i_0} \partial \tilde{x}_{i_0}}(0) = 4a_{i_0 j_0}(0) \neq 0$. The situation is then reduced to Case 1.

Since $a_{11}(0) \neq 0$ it is either always positive or always negative in some neighborhood of 0.

We define variable u_1 as:

$$u_1 = \sqrt{|a_{11}|} \left(x_1 + \frac{1}{|a_{11}|} \sum_{j=2}^n x_j a_{1j} \right)$$

$$u_i = x_i, \forall i \geq 2$$

Then we can rewrite f as:

$$f(x) = \sum_{j=1}^n \sum_{i=1}^n x_i x_j a_{ij} = \text{sgn}(a_{11}) u_1^2 + \sum_{j=2}^n \sum_{i=2}^n u_i u_j a'_{ij} = \text{sgn}(a_{11}) u_1^2 + \sum_{i,j=2}^n x_i x_j a'_{ij}$$

where a'_{ij} is smooth on some neighborhood of 0. In addition, by definition of u_i we have

$$\begin{aligned} \frac{\partial u_1}{\partial x_1}(0) &= \sqrt{|a_{11}(0)|} \neq 0 \\ \frac{\partial u_1}{\partial x_j}(0) &= \frac{a_{1j}(0)}{\sqrt{|a_{11}(0)|}}, \forall j \geq 2 \\ \frac{\partial u_i}{\partial x_j}(0) &= \delta_i^j \text{ otherwise.} \end{aligned}$$

Therefore, the Jacobian matrix $\partial u / \partial x$ at 0 is upper triangular, and its determinant is

$\sqrt{|a_{11}(0)|} > 0$. Shrinking the neighborhood again if necessary, by the Inverse Function Theorem $u_1, \dots, u_n$ define local coordinates near 0. Moreover we have

$$\det(a'_{ij}(0)) = \det(a_{ij}(0)) \left(\det \left(\frac{\partial u}{\partial x}(0) \right) \right)^2 \neq 0.$$

The rest follows by induction:

Suppose we have already constructed $u_1, \dots, u_{r-1}$ in a neighborhood U_{r-1} of 0, with

$$f = \pm(u_1)^2 \pm \dots \pm (u_{r-1})^2 + \sum_{i,j=r}^n x_i x_j b_{ij}(x_1, \dots, x_n).$$

By performing the same procedure as above, we may define u_r on a possibly smaller neighborhood, denoted as U_r , with

$$u_r = \sqrt{|b_{rr}|} \left(x_r + \frac{1}{|b_{rr}|} \sum_{j>r} x_j b_{jr} \right).$$

Then we have:

$$f = \pm(u_1)^2 \pm \dots \pm (u_r)^2 + \sum_{i,j=r+1}^n x_i x_j b'_{ij}.$$

By induction we have

$$f(x) = \pm u_1^2 \pm \dots \pm u_n^2 = - \sum_{i=1}^{\lambda} u_i^2 + \sum_{i=\lambda+1}^n u_i^2$$

(reordered if necessary) for some $0 \leq \lambda \leq n$. Under the new coordinates, the entries of the Hessian matrix are

$$\frac{\partial^2 f}{\partial u_i \partial u_j} = \begin{cases} -2 & \text{if } i = j \leq l \\ 2 & \text{if } i = j > l \\ 0 & \text{otherwise} \end{cases}$$

i.e. locally

$$Hf = \begin{pmatrix} -2 & & & & \\ & \ddots & & & \\ & & -2 & & \\ & & & 2 & \\ & & & & \ddots \\ & & & & & 2 \end{pmatrix}$$

There are exactly l negative entries and $n - l$ positive entries, so $l = \lambda$ is the index at p . $\square$

Also notice that, under the same basis $\{\frac{\partial}{\partial u_i}\}_{i=1}^n$ the differential df is

$$df = (-2u_1, \dots, -2u_\lambda, 2u_{\lambda+1}, \dots, 2u_n)$$

which is nonzero on some deleted neighborhood of the non-degenerate critical point $p = 0$. Hence non-degenerate critical points are isolated.

We use Morse's Lemma to study local properties of Morse functions. There are two cases: (1) near a regular point and (2) near a non-degenerate critical point.

Theorem 3.3 (Morse Theorem I). *Let f be smooth on M . Suppose $f^{-1}[a, b]$ is compact and contains no critical points for some closed interval $[a, b] \subset \mathbb{R}$. Then M^a is diffeomorphic to M^b and also a deformation retract of M^b .*

Proof. Equip M with a Riemannian metric g , such that the gradient ∇f is defined. Note that $\nabla f(p) = 0$ if p is a critical point of f . Let $\gamma : (-\epsilon, \epsilon) \rightarrow M$ be the integral curve of ∇f near p . Then we have

$$\left\langle \frac{d\gamma}{dt}, \nabla f \right\rangle = \frac{d}{dt} f \circ \gamma.$$

Define smooth function ρ on M , such that

$$\rho = \frac{1}{\|\nabla f\|^2}$$

on $f^{-1}([a, b])$, and that ρ vanishes outside a neighborhood of $f^{-1}([a, b])$. Consider the smooth vector field $X = \rho \nabla f$, which has compact support in M . We prove the following result as a lemma:

Theorem 3.4. *A smooth vector field on M which vanishes outside a compact set $K \subset M$ is complete.*

Proof. This is a corollary of the Fundamental Theorem on Flow and Uniform Time Lemma. Let X be a smooth vector field on M with compact support. Then at each point $p \in K$ there exists a neighborhood denoted as U_p and $\epsilon_p > 0$ such that the flow φ_t is defined for $(t, q) \in (-\epsilon_p, \epsilon_p) \times U_p$. Since K is compact, the open cover formed by U_p has a finite subcover. Therefore we may take ϵ as the smallest among all ϵ_p , and the composition of all φ_t gives a flow in K . Since $X = 0$ outside K , the integral curves outside K are constant. This shows that the flow is actually defined for all $t \in \mathbb{R}$. $\square$

By Theorem 3.4, there exists a unique flow diffeomorphism $\theta_t : M \rightarrow M$. Now for $q \in f^{-1}([a, b])$ we consider the function

$$\varphi : \mathbb{R} \rightarrow \mathbb{R}, \varphi(t) = f \circ \theta_t(q).$$

We have

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \varphi &= \left. \frac{d}{dt} \right|_{t=0} f \circ \theta_t(q) \\ &= \left\langle \left. \frac{d}{dt} \right|_{t=0} \theta_t(q), \nabla f \right\rangle \\ &= \langle X(q), \nabla f(q) \rangle = 1. \end{aligned}$$

Therefore, θ_{b-a} is a diffeomorphism from M^a to M^b . The homotopy equivalence is constructed similarly: define $r_t : [0, 1] \times M \rightarrow M$, with

$$r(t, q) = \begin{cases} q & \text{if } f(q) \leq a \\ \theta_{t(a-f(q))} & \text{if } a \leq f(q) \leq b \end{cases}$$

This is indeed a homotopy since

$$r(0, q) = q, r(1, q) = \theta_{a-f(q)}(q) \in M^a$$

for $q \in f^{-1}[a, b]$. The proof of Theorem 3.3 is then complete. $\square$

Theorem 3.5 (Morse Theorem II). *Let f be a smooth function on M . Suppose $f^{-1}[a, b]$ is compact and contains exactly one critical point p , which is non-degenerate with index λ . Then M^b is homotopy equivalent to $M^a \cup e^\lambda$, where e^λ is a λ -cell, i.e. a closed ball of dimension λ , with its boundary glued into M^a .*

Proof. By Theorem 3.3, M^b and M^a are homotopy equivalent outside a neighbourhood of p . Hence it suffices to show that the statement holds locally near p . WLOG we still assume $f(p) = 0$. By the Morse Lemma (Lemma 3.1), there exists a neighborhood U of p with a coordinate system $\{x_i\}_{i=1}^n$ such that

$$f = - \sum_{i=1}^{\lambda} x_i^2 + \sum_{i=\lambda+1}^n x_i^2$$

By shrinking to a smaller neighbourhood $V \subseteq U$ if necessary, we may assume (V, p) is

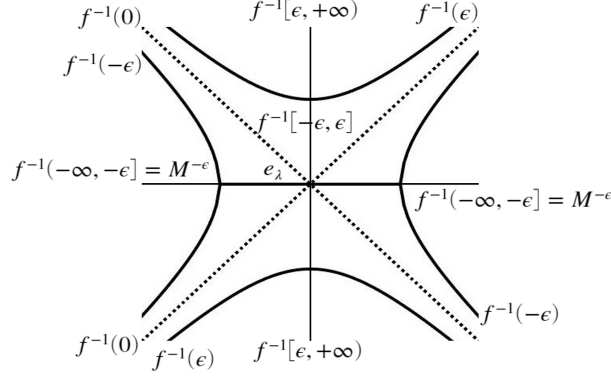


Figure 1: The horizontal axis denotes the coordinate subspace $x_1 = \dots = x_\lambda = 0$ and the vertical axis denotes $x_{\lambda+1} = \dots = x_n = 0$. The level sets $f^{-1}(\epsilon)$ and $f^{-1}(-\epsilon)$ are represented by two branches of hyperbola each. The asymptotes represent $f^{-1}(0)$. The horizontal line segment connecting the left/right branches is the λ -cell attached to $M^{-\epsilon}$ along its boundary.

diffeomorphic to $(\bar{V}, 0) \subseteq \mathbb{R}^n$ via $\varphi : V \rightarrow \mathbb{R}^n$ such that $\varphi(p) = 0$. The function on $\mathbb{R}^n$ would then be a composition of f with φ^{-1} . Choose $\epsilon > 0$, such that $f^{-1}(-r, r) \subset \bar{V}$ where $r = \max(2\epsilon, \sqrt{2}\epsilon)$. And in $\bar{V}$ exists a smaller closed n ball denote as:

$$\overline{B(0, r)} \subseteq \bar{V} \subseteq \mathbb{R}^n$$

Define an λ -cell in such a closed ball as:

$$e^\lambda = \overline{B^\lambda(0, \epsilon)} \times \{0\} = \{x \in \bar{V} | x = \{x_1, \dots, x_\lambda, 0 \dots, 0\}, \sum_{i=1}^{\lambda} x_i^2 \leq \epsilon\}$$

Notice that:

$$e^\lambda \cap M^{-\epsilon} = \{x \in \bar{V} | x = \{x_1, \dots, x_\lambda, 0 \dots, 0\}, \sum_{i=1}^{\lambda} x_i^2 = -\epsilon\} \subseteq \partial e^\lambda$$

proved the e^λ attached to sublevel set $M^{-\epsilon}$ is well-defined.

Now consider $\xi, \eta : \bar{V} \rightarrow [0, \infty)$ as $\xi = \sum_{i=1}^{\lambda} x_i^2$ and $\eta = \sum_{i=\lambda+1}^n x_i^2$. Then in $\bar{V}$ we have $f = -\xi + \eta$. Let $\mu : \mathbb{R} \rightarrow \mathbb{R}$ be a decreasing function such that:

$$\mu(0) > \epsilon, \mu(k) = 0 \quad \text{for } k \geq 2\epsilon$$

$$-1 \leq \mu'(k) \leq 0 \quad \forall k \in \mathbb{R}$$

Define a smooth function $F \in C^\infty(M)$ such that $F = f - \mu(\xi + 2\eta)$. Notice that for any point $q \in M \setminus V$ we have $F = f$ since μ vanished. Additionally, it is compact support in V

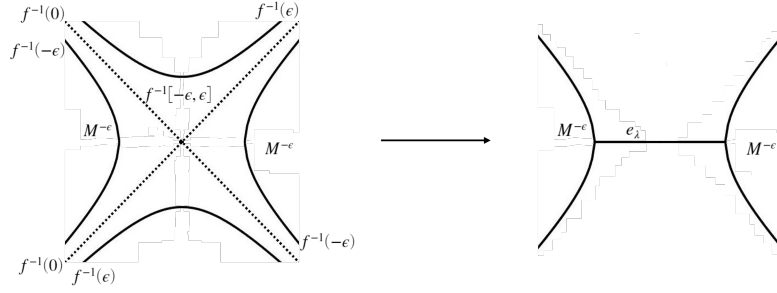


Figure 2: As shown in the graph, we aimed to deform M^ϵ to the $M^{-\epsilon}$ attached with λ -cell(horizontal line) we construct.

since:

$$\{q \in \bar{V} : -\xi(q) + 2\eta(q) \leq 2\epsilon\} \subset \{q \in \bar{V} : -\xi(q) + \eta(q) \leq 2\epsilon\} = \overline{B^n(x)} \subset \bar{V}$$

Therefore, F is a well-defined smooth function.

Assertion 1. The sublevel set $F^{-1}(-\infty, c + \epsilon) = M^\epsilon = f^{-1}(-\infty, c + \epsilon)$.

We already show that $F = f$ by definition on $M \setminus V$ and $F \leq f$ for any point $q \in V$. However we also have:

$$f = -\xi + \eta \leq \frac{1}{2}\xi + \eta \leq \epsilon$$

whenever $\xi(q) + 2\eta(q) \leq 2\epsilon$. Otherwise it will reduce to the case where $q \in M \setminus V$. Therefore it proves that $M^\epsilon = F^{-1}(-\infty, \epsilon)$.

Assertion 2. F and f has the same critical points.

Notice from the assumption of μ :

$$\frac{\partial F}{\partial \xi} = -1 - \mu'(\xi + 2\eta) < 0$$

$$\frac{\partial F}{\partial \eta} = 1 - 2\mu'(\xi + 2\eta) > 1$$

$$dF = \frac{\partial F}{\partial \xi} d\xi + \frac{\partial F}{\partial \eta} d\eta$$

Therefore from assumption the only case $dF = 0$ is $d\xi = d\eta = 0$ which is the critical point p .

Therefore, p is the only critical point in V and the critical point of f and F are same.

Assertion 3. $F^{-1}(-\infty, -\epsilon)$ is a deformation retract of M^ϵ .

Since $F \leq f$ implies that $M^{-\epsilon} \subseteq F^{-1}(-\infty, -\epsilon)$, which also implies that $F^{-1}(-\epsilon, \epsilon) \subset f^{-1}(-\epsilon, \epsilon)$. Let H denote for the closure of $F^{-1}(-\infty, -\epsilon) \setminus M^{-\epsilon}$. From the second assumption, there only contains critical point $p \in f^{-1}(-\epsilon, \epsilon)$. However, since $\mu(\xi(p) + 2\eta(p)) = \mu(0) > \epsilon$

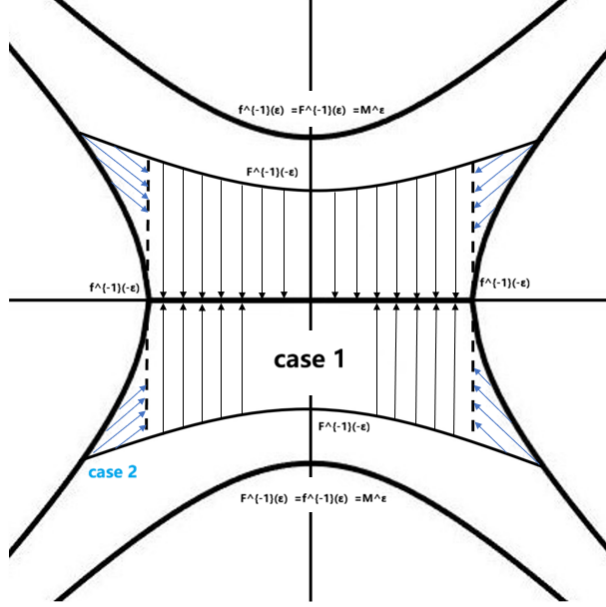


Figure 3: This denote for the cases we used to prove Assertion 4. We start with the case one which is the area with black arrows, then case two denote for all the blue arrows. Rest of the case three is the area under the hyperbola $f^{-1}(-\epsilon) = M^{-\epsilon}$

this implies that:

$$F(p) = f(p) - \mu(\xi(p) + 2\eta(p)) < -\epsilon$$

which shows that $p \in H$. Therefore, since there is no critical point in $M^\epsilon \setminus F^{-1}(-\infty, -\epsilon)$ from Morse Theorem I 3.3 $F^{-1}(-\infty, -\epsilon)$ is a deformation retract of M^ϵ .

Assertion 4. $M^{-\epsilon} \cup e^\lambda$ is a deformation retract of $M^{-\epsilon} \cup H$.

To construct a homotopy operator $r_t : M^{-\epsilon} \cup H \rightarrow M^{-\epsilon} \cup H$, we need to split it into three cases which separately been shown in Figure 3:

Case 1: When $\xi \leq \epsilon$, construct r_t as following:

$$r_t(x_1, \dots, x_n) = (x_1, \dots, x_\lambda, (1-t)x_{\lambda+1}, \dots, (1-t)x_n)$$

which is obvious that r_1 is the identity map and r_0 maps all the point in case 1 area in to the λ -cell. Recall that previously $\frac{\partial F}{\partial \eta} > 0$ has been proved, therefore, the map r_t is perfectly restricted in $M^{-\epsilon} \cup e^\lambda$.

Case 2: When $\epsilon \leq \xi \leq \eta + \epsilon$, construct r_t as following:

$$r_t = (x_1, \dots, x_\lambda, s_t x_{\lambda+1}, \dots, s_t x_n)$$

such that $s_t = t + (1-t)\sqrt{(\xi - \epsilon)/(\eta)}$. It is easily to check that $t = 1$ it is the identity map

and when $t = 0$ the map r_t will map the whole surface to the vertical line $\xi = \epsilon$ which can reduce to the Case 1.

Case 3: When $\xi \geq \eta + \epsilon$. Then r_t is just the identity map since that implies the point has already been contained in $M^{-\epsilon}$. The only case $\xi = \eta + \epsilon$ has also been considered and coincidence with Case 2.

Therefore, since from assertion 3 we have deformation retract from M^ϵ to $M^{-\epsilon} \cup H$ and from assertion 4 we have deformation retract from $M^{-\epsilon} \cup H$ to $M^{-\epsilon} \cup e^\lambda$, this complete the proof of Theorem 3.5. $\square$

Corollary 3.6. *Let M be a smooth manifold, and f be a strictly non-degenerate Morse function on M . Assuming that $f^{-1}[a, b]$ is compact for some closed and bounded interval $[a, b] \subset \mathbb{R}$ with a, b being regular values of f , we have*

$$M^b \sim M^a \cup_{g_1} e^{k_{p_1}} \cup_{g_2} \cdots \cup_{g_n} e^{k_{p_n}},$$

where $p_1, \dots, p_n \in M$ are the critical points of f that lie in $f^{-1}[a, b]$, with index k_{p_i} and $f(p_i)$ in increasing order, g_i are the gluing maps, and $\sim$ denotes homotopy equivalence. In particular, if M is compact, taking $a < \min f(M)$ and $b > \max f(M)$, we have $M^a = \emptyset$, $M^b = M$, and

$$M \sim e^{k_{p_1}} \cup \cdots \cup e^{k_{p_n}}$$

over all critical points in M .

Proof. The proof is just by applying Theorems 3.3 and 3.5 repeatedly, noting that the union is finite, because there are at most finitely (non-degenerate) critical points, as M is assumed to be compact. $\square$

4 Existence of Morse Functions

Corollary 3.6 implies that the geometry of a smooth manifold is, up to homotopy equivalence, determined by the indices of a Morse function near its critical points. This allows us to transform the study of manifolds into studying local properties of such Morse functions. However, before applying the result, a natural question is whether such functions exist or not. In fact, this is guaranteed by the following theorem, which we state without proof.

Theorem 4.1 (Morse-Thom). *The collection of all Morse functions on a smooth manifold M is dense in $C^\infty(M)$. The collection of all strictly non-degenerate Morse functions on M is also dense in $C^\infty(M)$.*

The result is highly non-trivial, and we refer the readers to [1] for a proof. Instead we present here two constructions of a particular class of Morse functions. Note that every smooth manifold can be embedded into the Euclidean space, by the Whitney Embedding Theorem (Theorem 6.15 of [4]). Hence it suffices to consider M as an embedded submanifold in $\mathbb{R}^n$.

Lemma 4.2. *Let $M \subset \mathbb{R}^n$ be a smooth embedded submanifold. For each $a = (a_1, \dots, a_n) \in \mathbb{R}^n$, define*

$$f_a : M \rightarrow \mathbb{R}, x \mapsto a \cdot x = \sum_{i=1}^n a_i x_i. \quad (4.1)$$

(Note that f_a is the restriction of a linear functional on $\mathbb{R}^n$, and hence is smooth.) Then f_a is a strictly non-degenerate Morse function for $a \in \mathbb{R}^n$ almost everywhere.

Proof. (See also Proposition 17.18 in [2].) For each $p \in M$, let $U_p \subset M$ be a chart neighbourhood with local coordinates $y_1, \dots, y_m$. Since M is 2nd countable, the open cover $\{U_p\}_{p \in M}$ has a countable subcover $\{U_i\}$. On each U_i , we consider the gradient $g(y) = \nabla_y f = (\sum_i a_i \partial_{y_j} x_i)_j$. Note that a point $p \in U_i$ is a critical point iff $\nabla_y f(p) = g(y)|_p = 0$, and non-degenerate iff $Hf(p) = (\partial_{y_k} g_j|_p)_{j,k}$ is a non-singular $m \times m$ matrix at p . On the other hand, since M is a submanifold in $\mathbb{R}^n$ of dimension m , the n vectors $\nabla_y x_i|_p$ span the tangent space $T_p M \subset \mathbb{R}^n$, and we may choose m of them, say $\nabla_y x_1|_p, \dots, \nabla_y x_m|_p$ without loss of generality, that form a basis of it. By the Rank Theorem (Theorem 4.12 of [4]), there is a neighbourhood $V \subset U_i$ of p on which the Jacobian matrix $J(q) = (\partial_{y_j} x_i|_q)_{i,j=1}^m$ is invertible at each $q \in V$. We fix $a_{m+1}, \dots, a_n \in \mathbb{R}$, and define

$$h_{a_{m+1}, \dots, a_n} : V \rightarrow \mathbb{R}^{n-m}, q \mapsto J(q)^{-1} \left(\sum_{i=m+1}^n a_i \frac{\partial x_i}{\partial y_j}(q) \right)_{j=1}^m.$$

Hence, for $a \in \mathbb{R}^n$, $p \in U_i$ is a critical point of f_a iff $h_{a_{m+1}, \dots, a_n}(p) = (a_1, \dots, a_m)$, and p is non-degenerate iff $(a_1, \dots, a_m)$ is a regular value of $h_{a_{m+1}, \dots, a_n}$. By Sard's Theorem (Theorem 6.10 of [4]), for each fixed choice of $(a_{m+1}, \dots, a_n)$, such $(a_1, \dots, a_m)$ exists almost everywhere in $\mathbb{R}^m$. Hence also $a = (a_1, \dots, a_n)$ exists almost everywhere in $\mathbb{R}^n$. In fact we may further require f to be strictly non-degenerate: note that there are at most countably many critical points $p \in M$, and the choice of a such that any two of such critical points are mapped to the same value corresponds to a countable union of vector subspaces in $\mathbb{R}^n$ of strictly positive codimension, which is still of measure zero. Hence $f = f_a$ is a strictly non-degenerate Morse function for a almost everywhere. $\square$

Another approach is presented in [5]. Consider the squared distance function $L_a : M \rightarrow \mathbb{R}, p \mapsto \|a - p\|^2$, where $a \in \mathbb{R}^n$ is a fixed choice of point. It is clear that L_a is a smooth function. Moreover, we have the following:

Lemma 4.3. *Let $M \subset \mathbb{R}^n$ be a smooth submanifold. For $a \in \mathbb{R}^n$ define the squared distance function*

$$L_a : M \rightarrow \mathbb{R}, p \mapsto \|a - p\|^2 = \sum_{i=1}^n (a_i - p_i)^2.$$

Then L_a is a strictly non-degenerate Morse function for $a \in \mathbb{R}^n$ almost everywhere.

To show Lemma 4.3 we first prove the following. Lemma 4.3 will then immediately follow from Sard's Theorem.

Lemma 4.4. *Let $M \subset \mathbb{R}^n$ be a smooth submanifold, and NM be its normal bundle. Note that NM is an n -dimensional submanifold in $\mathbb{R}^{2n}$. Define $E : NM \rightarrow \mathbb{R}^n, (p, v) \mapsto p + v$. Then:*

- (1) *A point $p \in M$ is a critical point of L_a if and only if $(p, a - p)$ lies in NM .*
- (2) *In the case of (1), p is non-degenerate if and only if $(p, a - p)$ is a regular point of E .*

Proof. We choose local coordinates $(y_1, \dots, y_m)$ of M near p . Then the gradient of L_a is $\nabla_y L_a(p) = -2J(p)(a - p)$, where $J(p) = (\partial_{y_i} x_j)$ is the $m \times n$ Jacobian matrix at p . Since $J(p)$ is always of rank m , it maps $\mathbb{R}^n$ onto $T_p M$, so p is a critical point of L_a iff $a - p \in (T_p M)^\perp = N_p M$. Hence (1) follows.

Writing the Hessian of L_a as $HL_a(p) = (h_{ij}(p))$, we have

$$\frac{1}{2} h_{ij}(p) = \sum_{k=1}^n \partial_{y_i} x_k \partial_{y_j} x_k - (a_k - x_k) \partial_{y_i} \partial_{y_j} x_k = \langle \partial_{y_i} x, \partial_{y_j} x \rangle - \langle a - p, \partial_{y_i} \partial_{y_j} x \rangle.$$

On the other hand, we may choose coordinates $(t_1, \dots, t_{n-m})$ in the normal direction, associated with some local frame $w_1, \dots, w_{n-m}$ of NM near p . Under the coordinates (y, t) , the map E can be written as

$$E : (y_1, \dots, y_m, t_1, \dots, t_{n-m}) \mapsto x(y) + \sum_{i=1}^{n-m} t_i w_i(y),$$

and hence its differential $dE : T(NM) \rightarrow T(NM)$ is represented by the $n \times n$ matrix

$$dE = \begin{pmatrix} (\partial_{y_i} x^T + \sum_{\ell=1}^{n-m} t_\ell \partial_{y_i} w_\ell^T)_{1 \leq i \leq m} \\ w_1^T \\ w_2^T \\ \dots \\ w_{n-m}^T \end{pmatrix}$$

in the standard basis $\{\partial_{y_1} x, \dots, \partial_{y_m} x, w_1, \dots, w_{n-m}\}$. On the other hand, the $n \times n$ matrix

$$B = \begin{pmatrix} \partial_{y_1} x & \dots & \partial_{y_m} x & w_1 & \dots & w_{n-m} \end{pmatrix}$$

is non-singular, so the matrix

$$(dE)B = \begin{pmatrix} (\langle \partial_{y_i} x, \partial_{y_j} x \rangle + \sum_{\ell=1}^{n-m} t_\ell \langle \partial_{y_i} w_\ell, \partial_{y_j} x \rangle)_{i,j=1}^m & (\sum_{\ell=1}^{n-m} \langle \partial_{y_i} w_\ell, w_j \rangle)_{1 \leq i \leq m, 1 \leq j \leq n-m} \\ O_{(n-m) \times m} & (w_i^T w_j)_{i,j=1}^{n-m} \end{pmatrix}$$

has the same rank as dE . Note that the lower-right block is of full rank $n - m$ because as $w_1, \dots, w_{n-m}$ are linearly independent, and the lower-left block is zero since $\partial_{y_i} x$ are tangential and w_j are normal to M . Hence dE is non-singular iff the upper-left block, which we denote by $C = (c_{ij})$, is so. Now notice that

$$\partial_{y_i} \langle w_\ell, \partial_{y_j} x \rangle = \langle \partial_{y_i} w_\ell, \partial_{y_j} x \rangle + \langle w_\ell, \partial_{y_i} \partial_{y_j} x \rangle = 0,$$

so we have

$$c_{ij} = \langle \partial_{y_i} x, \partial_{y_j} x \rangle - \left\langle \sum_{\ell=1}^{n-m} t_\ell w_\ell, \partial_{y_i} \partial_{y_j} x \right\rangle.$$

In particular, at $(p, a - p)$, we have

$$c_{ij}(p, a - p) = \langle \partial_{y_i} x|_p, \partial_{y_j} x|_p \rangle - \langle a - p, \partial_{y_i} \partial_{y_j} x|_p \rangle = \frac{1}{2} h_{ij}(p).$$

Hence p is non-degenerate iff $2C(p, a - p) = HL_a(p)$ is non-singular, i.e. $(p, a - p)$ is a regular point of E . $\square$

Remark 4.5. The vector valued matrix defined in local coordinates as

$$A := ((\partial_{y_i} \partial_{y_j} x)^N)_{i,j}$$

on an embedded submanifold $M \subset \overline{M}$ in a Riemannian manifold is called the *second*

fundamental form, where N denotes the normal component (i.e. projection onto the normal space). One can see that it is independent of the choice of coordinates as a bilinear map

$$A : T_p M \times T_p M \rightarrow N_p M, (\partial_{y_i}, \partial_{y_j}) \mapsto (\partial_{y_i} \partial_{y_j} x)^N.$$

Proof of Lemma 4.3. By Lemma 4.4, L_a has a degenerate critical point iff a is a critical value of E , and by Sard's Theorem, such points form a measure zero set in $\mathbb{R}^n$. The existence of strictly non-degenerate L_a follows from a very similar argument as in the proof of Lemma 4.2, noting that the collection of $a \in \mathbb{R}^n$ such that $L_a(p) = L_a(q)$ for distinct $p, q \in M$ form a hyperplane in $\mathbb{R}^n$ which is again measure zero. $\square$

Thus, for any compact smooth manifold M , we can indeed find a strictly non-degenerate Morse function on it, so that it can be reconstructed (up to homotopy) by gluing cells successively as in Corollary 3.6, starting from the unique minimum that corresponds to a 0-cell. We note that this is very close to the definition of a CW complex:

Definition 4.6 (CW complex). A (finite) *CW complex* X is a topological space obtained by gluing a finite collection of cells successively with increasing dimension, starting from a finite set of points X^0 . More precisely, at step k one glues a finite collection of k -cells e_i^k to X^{k-1} via continuous maps $g_i^k : \partial e_i^k = \mathbb{S}^{k-1} \rightarrow X^{k-1}$ (called the *gluing maps*), obtaining X^k . If X is a CW complex, we denote by X^k the subspace containing all cells of dimension $i \leq k$ (it is called the *k-skeleton* of X). The *dimension* of X is the largest dimension of the cells in it.

In fact, the above observation is indeed correct. We have the following:

Proposition 4.7. *Every compact smooth manifold M has the same homotopy type with some CW complex X .*

Proof. By Lemmas 4.2 or 4.3, strictly non-degenerate Morse functions always exist. This allows us to construct a topological space $\tilde{X}$ through gluing cells as in Corollary 3.6. It remains to show that $\tilde{X}$ is homotopy equivalent to a CW complex X , i.e. the cells can be re-glued according to their dimensions. Suppose at some step a k -cell is glued to $\tilde{X}^n$ via $g : \mathbb{S}^{k-1} \rightarrow \tilde{X}^n$, with $k \leq n$. Since $\mathbb{S}^{k-1}$ is of strictly lower dimension than an n -cell, for any n -cell e^n in $\tilde{X}^n$ that intersects the image of g , their intersection $e^n \cap g(\mathbb{S}^{k-1})$ is properly contained in e^n . It is then a general fact in topology that any closed ball $e^n = \overline{B^n}$ minus an interior point can be contracted to its boundary $\mathbb{S}^{n-1}$. In our case, this implies that $e^n \cap g(\mathbb{S}^{k-1})$ can be deformed into a subset in the boundary of e^n , which by assumption is glued to $\tilde{X}^{n-1}$. Doing this for all the n -cells in $\tilde{X}^n$, we deform the image of g into $\tilde{X}^{n-1}$. If

$k \leq n - 1$, we may perform the above procedure repeatedly until the image of g is contained in $\tilde{X}^{k-1}$, and we obtain X^k . The resulting space X is a CW complex. $\square$

One reason of using CW complexes as a model for manifolds is that their geometry is more simple, in the sense that their (co)homology are easier to compute, as we will see in the following section.

5 Results from Differential and Algebraic Topology

Before stating the results, we first recall some definitions and theorems on homology and cohomology. The main references for this section are [2, 4].

Definition 5.1 (De Rham cohomology). Let M be a smooth manifold, and denote by $\Omega^k(M)$ the space of smooth k -forms on M . Note that $\Omega^\bullet(M)$ is a cochain complex under the de Rham differential d (i.e. $d^2 = 0$); if we denote

$$\begin{aligned} Z^k(M) &:= \ker(d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)) \subset \Omega^k(M) \\ B^k(M) &:= \operatorname{im}(d : \Omega^{k-1}(M) \rightarrow \Omega^k(M)) \subset \Omega^k(M) \end{aligned}$$

the subspace of closed and exact k -forms respectively, this means $B^k(M) \subset Z^k(M)$, so we may define the corresponding cohomology $H^\bullet(M)$, with

$$H^k(M) := Z^k(M) / B^k(M).$$

It is called the *de Rham cohomology*. In particular the degree 0 part $H^0(M)$ counts the number of connected components of M , i.e. $H^0(M) = \mathbb{R}^m$ if M has m connected components. The cohomology is a graded ring under the *cup product*: if $[\alpha] \in H^p(M)$, $[\beta] \in H^q(M)$, then

$$[\alpha] \cup [\beta] := [\alpha \wedge \beta] \in H^{p+q}(M).$$

Note that it is well-defined since the wedge product behaves well under d : if α, β are both closed, then so is $\alpha \wedge \beta$; if in addition one of them is exact, then so is $\alpha \wedge \beta$. Moreover, if M, N are smooth manifolds and $f : M \rightarrow N$ is a smooth map, then the pull-back $f^* : \Omega^\bullet(N) \rightarrow \Omega^\bullet(M)$ is a degree-preserving chain map (i.e. it intertwines the differentials, with $f^* \circ d_N = d_M \circ f^*$), so it descends to a map $f^* : H^\bullet(N) \rightarrow H^\bullet(M)$ on the cohomology level.

The de Rham cohomology is homotopy invariant, i.e.:

Lemma 5.2. *Let M, N be smooth manifolds, and $f, g : M \rightarrow N$ be smooth maps that are homotopic. Then they induce the same map $f^* = g^* : H^\bullet(N) \rightarrow H^\bullet(M)$. Consequently, if M, N are homotopy equivalent, then they have isomorphic cohomology rings.*

The proof is by constructing a *homotopy operator*:

Definition 5.3. Let M, N be smooth manifolds. A *homotopy operator* between two smooth maps $f, g : M \rightarrow N$ is a (linear) map $Q : \Omega^\bullet(N) \rightarrow \Omega^{\bullet-1}(M)$ of degree -1 , satisfying $f^* - g^* = dQ + Qd$. Note that in this case $f^* = g^*$ on the cohomology level: we have $f^*[\alpha] = g^*[\alpha] + [dQ\alpha] + [Qd\alpha] = g^*[\alpha]$ for $[\alpha] \in H^\bullet(N)$ represented by the closed form α .

Proof of Lemma 5.2. It suffices to show that a homotopy operator between f, g exists if they are homotopic. Let $H : M \times [0, 1] \rightarrow N$ be a homotopy between f and g , i.e. $f(p) = H(p, 0)$ and $g(p) = H(p, 1)$ for all $p \in M$. Let $W := M \times \mathbb{R}$, and consider the vector field $X = \partial_t$ on W , where t is the coordinate in $\mathbb{R}$. Then X is complete, with flow

$$\varphi_t : W \rightarrow W, (p, a) \rightarrow (p, a + t)$$

which exists for all $t \in \mathbb{R}$. Define

$$\tilde{Q}_t = \iota_X \circ \varphi_t^* : \Omega^k(W) \rightarrow \Omega^{k-1}(W); \quad \tilde{Q} : \Omega^k(W) \rightarrow \Omega^{k-1}(W), \tilde{Q}(\omega) = \int_0^1 \tilde{Q}_t(\omega) dt.$$

Here ι_X denotes contracting with the vector field X . By definition $\tilde{Q}_t$ and hence $\tilde{Q}$ are $\mathbb{R}$ -linear. Moreover for $\omega \in \Omega^k(W)$ we have

$$\begin{aligned} d\tilde{Q}(\omega) + \tilde{Q}(d\omega) &= \int_0^1 (d\iota_X \varphi_s^* \omega + \iota_X \varphi_s^* d\omega) ds \\ &= \int_0^1 (d\iota_X \varphi_s^* \omega + \iota_X d\varphi_s^* \omega) ds \\ &= \int_0^1 \mathcal{L}_X \varphi_s^* \omega ds \\ &= \int_0^1 \left. \frac{d}{dt} \right|_{t=0} \varphi_t^* \varphi_s^* \omega ds \\ &= \int_0^1 \left. \frac{d}{dt} \right|_{t=s} \varphi_t^* \omega ds = \varphi_1^* \omega - \omega. \end{aligned}$$

Here on the 3rd line we used Cartan's Formula (Theorem 14.35 in [4]). Let $i : M \rightarrow W$ be the inclusion of M into W as the fibre over 0, i.e. $i(p) = (p, 0)$. Then $f = H \circ i$ and

$g = H \circ \varphi_1 \circ i$, so

$$g^* - f^* = i^* \circ (\varphi_1^* - \text{id}) \circ H^* = i^* \circ (d\tilde{Q} + \tilde{Q}d) \circ H^* = d(i^* \circ \tilde{Q} \circ H^*) + (i^* \circ \tilde{Q} \circ H^*)d.$$

The proof is then finished by taking $Q = i^* \circ \tilde{Q} \circ H^* : \Omega^k(N) \rightarrow \Omega^{k-1}(M)$. $\square$

Example 5.4. Suppose the manifold M is contractible, i.e. it is homotopy equivalent to a point. Then $H^0(M) = \mathbb{R}$ and $H^k(M) = 0$ for all $k > 0$. In the case $M = \mathbb{R}^n$, this is the Poincaré's Lemma.

The following result provides a very useful tool in computing the cohomology of manifolds.

Proposition 5.5 (Mayer-Vietoris Theorem). *Let M be a smooth manifold. Suppose it can be covered by two open sets $U, V \subset M$. Then there is a long exact sequence, called the Mayer-Vietoris sequence:*

$$\cdots \longrightarrow H^k(M) \longrightarrow H^k(U) \oplus H^k(V) \longrightarrow H^k(U \cap V) \longrightarrow H^{k+1}(M) \longrightarrow \cdots .$$

Proof. For each k , we define maps

$$\begin{aligned} i_k &= (i_U^*, i_V^*) : \Omega^k(M) \rightarrow \Omega^k(U) \oplus \Omega^k(V), \omega \mapsto (\omega|_U, \omega|_V); \\ j_k &: \Omega^k(U) \oplus \Omega^k(V) \rightarrow \Omega^k(U \cap V), (\alpha, \beta) \mapsto \alpha|_{U \cap V} - \beta|_{U \cap V}. \end{aligned}$$

(Note that the restriction maps $\omega \mapsto \omega|_U$ resp. $\omega|_V$ is pull-back by the inclusion $i_U : U \hookrightarrow M$ resp. $i_V : V \hookrightarrow M$.) Then the sequence of cochain complexes is exact:

$$0 \longrightarrow \Omega^\bullet(M) \longrightarrow \Omega^\bullet(U) \oplus \Omega^\bullet(V) \longrightarrow \Omega^\bullet(U \cap V) \longrightarrow 0.$$

(It is clear that i_k is injective and that $j_k \circ i_k = 0$. Let $\omega \in \Omega^k(U \cap V)$. Choose a partition of unity $\{\rho_U, \rho_V\}$ subordinate to the open cover $\{U, V\}$, and define $\omega_U \in \Omega^k(U)$ as $\omega_U = \rho_V \omega$ on $U \cap V$ and 0 elsewhere; and similarly $\omega_V \in \Omega^k(V)$ as $\omega_V = -\rho_U \omega$ on $U \cap V$ and 0 elsewhere. Then ω_U, ω_V are well defined k -forms, with $\omega = j_k(\omega_U, \omega_V)$. Hence j_k is surjective. Also, if $\alpha \in \Omega^k(U), \beta \in \Omega^k(V)$ agree on intersection, then they can be patched together as a global k -form $\omega \in \Omega^k(M)$; this shows $\ker j_k = \text{im } i_k$.) Then it is a general fact in homological algebra that such a short exact sequence of cochain complexes induces a long exact sequence of cohomology:

$$\cdots \longrightarrow H^k(M) \longrightarrow H^k(U) \oplus H^k(V) \longrightarrow H^k(U \cap V) \longrightarrow H^{k+1}(M) \longrightarrow \cdots .$$

□

As an application of the Mayer-Vietoris exact sequence, we compute the cohomology of the sphere $\mathbb{S}^n$.

Example 5.6. Consider $M = \mathbb{S}^n$. Let $U, V \subset M$ be the two chart domains corresponding to the stereographic projection, i.e. U, V are homeomorphic to $\mathbb{R}^n$. Moreover $U \cap V$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$, and hence $\mathbb{S}^{n-1} \times \mathbb{R}$, which is homotopy equivalent to $\mathbb{S}^{n-1}$ since $\mathbb{R}$ is contractible. The Mayer-Vietoris sequence for M is (when $k > 0$)

$$\cdots \longrightarrow H^k(\mathbb{S}^n) \longrightarrow H^k(\mathbb{R}^n) \oplus H^k(\mathbb{R}^n) = 0 \longrightarrow H^k(\mathbb{S}^{n-1}) \longrightarrow H^{k+1}(\mathbb{S}^n) \longrightarrow \cdots .$$

In particular, in the case $n = 1$, we have

$$0 \longrightarrow H^0(\mathbb{S}^1) = \mathbb{R} \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \longrightarrow H^1(\mathbb{S}^1) \longrightarrow 0$$

so $H^1(\mathbb{S}^1) = \mathbb{R}$. By induction on n , we have

$$H^k(\mathbb{S}^n) = \begin{cases} \mathbb{R} & \text{if } k = 0 \text{ or } n \\ 0 & \text{otherwise.} \end{cases}$$

The homology can be viewed as a “dual” version of the cohomology of manifolds. More generally, we define it for topological spaces:

Definition 5.7 (Singular homology). A (standard) n -simplex is

$$\Delta_n = \left\{ \sum_{i=1}^n t_i e_i : t_i \in \mathbb{R}_{\geq 0}, \sum_{i=1}^n t_i \leq 1 \right\}$$

in some $\mathbb{R}^m$ with $m \geq n$, where e_i is the i -th coordinate vector in $\mathbb{R}^m$. Alternatively, it is the convex polytope spanned with $n + 1$ vertices $e_0 = 0, \dots, e_n$, so we may also write

$$\Delta_n = \left\{ \sum_{i=0}^n t_i e_i : 0 \leq t_i \leq 1, \sum_{i=0}^n t_i = 1 \right\}.$$

A *singular n -simplex* in a topological space X is a continuous map $s : \Delta_n \rightarrow X$. The collection of all formal combinations of singular n -simplexes in X over a commutative ring R (with unity, as we always assume) is denoted $S_n(X, R)$ (its elements are called *singular n -chains* with *coefficient* in R), with $S_\bullet(X, R) := \bigoplus_{n \geq 0} S_n(X, R)$ the *singular chain complex*.

We define the i -th *face map* on Δ_n , $0 \leq i \leq n$, as

$$\partial^i : \Delta_{n-1} \rightarrow \Delta_n, (t_0, \dots, t_{n-1}) \mapsto (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{n-1}),$$

and the *boundary map*

$$\partial : S_n(X, R) \rightarrow S_{n-1}(X, R), s \mapsto \sum_{i=0}^n (-1)^i s \circ \partial^i,$$

extended R -linearly. Then $(S_\bullet(X, R), \partial)$ is a chain complex (i.e. $\partial^2 = 0$). If we denote

$$\begin{aligned} Z_k &:= \ker(d : S_k(X, R) \rightarrow S_{k-1}(X, R)) \subset S_k(X, R) \\ B_k &:= \text{im}(d : S_{k+1}(X, R) \rightarrow S_k(X, R)) \subset S_k(X, R) \end{aligned}$$

this means $B_k \subset Z_k$, so similar as before we may define $H_\bullet(X, R)$, which is a graded R -module, with

$$H_k(X, R) := Z_k / B_k.$$

It is called the *singular homology* with coefficients in R .

If $f : X \rightarrow Y$ is a continuous map between topological spaces, then it induces a degree preserving R -linear map on the chain level (called *push-forward* by f) $f_* : S_\bullet(X, R) \rightarrow S_\bullet(Y, R)$, $s \mapsto f \circ s$. By definition it satisfies $f \circ \partial_X = \partial_Y \circ f$, so it descends to the homology level as $f_* : H_\bullet(X, R) \rightarrow H_\bullet(Y, R)$.

Similar as before, homotopic maps induce the same map on the homology level. We also have the Mayer-Vietoris sequence for homology:

Proposition 5.8 (Mayer-Vietoris Theorem for homology). *Let X be a topological space, and $U, V \subset X$ be open sets that cover X . Then the sequence is exact:*

$$\dots \longrightarrow H_k(U \cap V) \longrightarrow H_k(U) \oplus H_k(V) \longrightarrow H_k(X) \longrightarrow H_{k-1}(U \cap V) \longrightarrow \dots$$

The dual version of the singular homology is the singular cohomology:

Definition 5.9 (Singular cohomology). Let X be a topological space. The *singular cochain complex* with coefficients in R is the dual R -module of the singular homology, i.e.

$$S^n(X, R) := \text{Hom}_R(S_n(X, R), R), \quad S^\bullet(X, R) := \bigoplus_{n \geq 0} S^n(X, R).$$

It has a differential δ that is the formal dual of ∂ , i.e.

$$\delta : S^n(X, R) \rightarrow S^{n+1}(X, R), \omega \mapsto \delta\omega, (\delta\omega)(c) = \omega(\partial c)$$

for any $c \in S_{n+1}(X, R)$. Similar as above, we have $\delta^2 = 0$, making it a cochain complex, and the corresponding cohomology is the *singular cohomology*, denoted $H^\bullet(X, R)$.

The next result shows that the two versions of cohomology coincide when the space is a smooth manifold and the coefficient ring is $\mathbb{R}$:

Theorem 5.10 (De Rham, Theorem 18.14 of [4]). *Let M be a smooth manifold. Then $H^\bullet(M) \simeq H^\bullet(M, \mathbb{R})$.*

The proof of this theorem is long and not so related to the topic of this paper, so we refer the readers to Section 18.3-18.4 of [4] for a proof.

For completeness here we include the following Universal Coefficient Theorem, which we state without proof.

Theorem 5.11 (Universal Coefficient Theorem, Theorem 15.14 of [2]). *Let X be a topological space, and R be a commutative ring. Then we have:*

$$\begin{aligned} H_n(X, R) &\simeq (H_n(X, \mathbb{Z}) \otimes_{\mathbb{Z}} R) \oplus \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, \mathbb{Z}), R); \\ H^n(X, R) &\simeq \text{Hom}_{\mathbb{Z}}(H_n(X, \mathbb{Z}), R) \oplus \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X, \mathbb{Z}), R). \end{aligned}$$

Here Tor is the (left) derived functor of the right exact functor $- \otimes_{\mathbb{Z}} R$, and Ext is the (right) derived functor of the left exact functor $\text{Hom}_{\mathbb{Z}}(-, R)$. For simplicity, in the next section we will work with coefficients in $\mathbb{R}$.

The singular (co)homology provides a way for us to understand the geometry of singular spaces, including CW complexes. One example for this is Morse's Inequalities, as stated below. First we define relative homology:

Definition 5.12. Let X be a topological space, and $A \subset X$ be a subspace. Note that the inclusion $i : A \rightarrow X$ induces an injective map $i_* : S_\bullet(A, R) \rightarrow S_\bullet(X, R)$. Consider the quotient

$$S_k(X, A, R) := S_k(X, R) / S_k(A, R).$$

Since the boundary map ∂ commutes with i_* , it descends to a well-defined map on $S_\bullet(X, A, R)$ which we still denote by ∂ and which still satisfies $\partial^2 = 0$. The corresponding homology $H_\bullet(X, A, R)$ is called the (k -th) *relative homology* of X with respect to A (with coefficient in

R). Similar as before, the short exact sequence of chain complexes

$$0 \longrightarrow S_{\bullet}(A, R) \longrightarrow S_{\bullet}(X, R) \longrightarrow S_{\bullet}(X, A, R) \longrightarrow 0$$

induces a long exact sequence in homology, called the *relative homology exact sequence*:

$$\cdots \longrightarrow H_k(A, R) \longrightarrow H_k(X, R) \longrightarrow H_k(X, A, R) \longrightarrow H_{k-1}(A, R) \longrightarrow \cdots$$

When the space X is a CW complex, the results are more simple (see also [3]). Here we omit the coefficient ring R .

Lemma 5.13 (Lemma 2.34 of [3]). *Let X be a finite CW complex of dimension n . Then:*

- (1) *The relative homology group $H_k(X^m, X^{m-1})$ is zero if $k \neq m$, and a free module of rank equal to the number of m -cells in X if $k = m$.*
- (2) *$H_k(X^m) = 0$ if $k > m$. Also $H_k(X^m) \simeq H_k(X)$ if $k < m$.*
- (3) *There is a chain complex*

$$H_n(X) = H_n(X^n, X^{n-1}) \longrightarrow H_{n-1}(X^{n-1}, X^{n-2}) \longrightarrow \cdots \longrightarrow H_0(X_0) \longrightarrow 0$$

where the maps are compositions $\partial : H_k(X^k, X^{k-1}) \rightarrow H_{k-1}(X^{k-1}) \rightarrow H_{k-1}(X^{k-1}, X^{k-2})$, satisfying $\partial^2 = 0$. It is called the *cellular chain complex*.

- (4) *The cellular chain complex induces the same homology as the usual homology $H_{\bullet}(X)$.*

In particular, if the coefficient ring R is Noetherian, all the homology $H_k(X)$ would be finitely-generated. In view of Proposition 4.7, de Rham's Theorem and the Universal Coefficient Theorem (Theorems 5.10 and 5.11), this shows the following:

Proposition 5.14. *Let M be a compact smooth manifold. Then its de Rham cohomology $H^{\bullet}(M)$ is a finite dimensional graded vector space.*

The following result, the Morse's Inequalities, is originally stated for CW complexes. On the other hand, by Proposition 4.7, we see that it also holds for smooth manifolds.

Theorem 5.15 (Morse's Inequalities). *Let M be smooth, with strictly non-degenerate Morse function f on M . Denote by c_k the number of critical points of f with index k , which is also the rank of $H_k(X^k, X^{k-1})$ if X is the CW complex homotopy equivalent to M . We also let b_k be the rank of $H_k(M) \simeq H_k(X)$. Then we have*

$$\sum_{i=0}^k (-1)^i c_{k-i} \geq \sum_{i=0}^k (-1)^i b_{k-i}$$

for each $0 \leq k \leq n$.

Proof. For simplicity we denote $C_k := H_k(X^k, X^{k-1})$, $H_k := H_k(X)$, and let $Z_k, B_k \subset C_k$ be the collection of cycles and boundaries in the cellular chain complex, with rank r_k and s_k resp. Note that $H_k = Z_k/B_k$, so we have two short exact sequences:

$$\begin{aligned} 0 \longrightarrow Z_k \longrightarrow C_k \longrightarrow B_{k-1} \longrightarrow 0 \\ 0 \longrightarrow B_k \longrightarrow Z_k \longrightarrow H_k \longrightarrow 0. \end{aligned}$$

Hence

$$c_k = r_k + s_{k-1} = b_k + s_k + s_{k-1}$$

for $1 \leq k \leq n$. For $k = 0$ we have

$$c_0 = r_0 = b_0 + s_0 \geq b_0.$$

This is the 1st (or 0th) Morse Inequality. Since $(c_1 - b_1) - (c_0 - b_0) = s_1 \geq 0$, we have

$$c_1 - c_0 \geq b_1 - b_0$$

which is the 2nd (or 1st) Morse Inequality. Repeating this process we may obtain the remaining inequalities. $\square$

6 Applications

One aspect of Morse theory is how the global geometry of a manifold is determined by the local properties of a suitably chosen function near its critical points. We start with the following example.

Proposition 6.1 (Reeb's Theorem). *Let M be a compact smooth manifold, and f be a smooth function on M . Suppose f has only 2 critical points, both non-degenerate. Then M is homotopy equivalent to $\mathbb{S}^n$.*

Proof. Since M is compact, the two critical points p, q must be the (unique) minimum point and maximum point of f . Since f attains minimum at p , Hf is positive definite at p , and $\text{index}_p f = 0$. Similarly, Hf is negative definite at q , so $\text{index}_q f = n$. So $M \sim e^0 \cup e^n$, and since the 0-cell e^0 is a point, the identification is obtained by identifying all the boundary of e^n to a point. So the resulting manifold is $\mathbb{S}^n$. $\square$

Below we demonstrate how Morse theory can be used to compute the (co)homology of manifolds. We start with the following lemma:

Lemma 6.2 (Homology of gluing spheres). *Let X be a path-connected topological space, and $Y = X \cup e^n$ be obtained by gluing the n -cell e^n to X , with $n > 1$, along its boundary $\mathbb{S}^{n-1}$, via a continuous map $g : \mathbb{S}^{n-1} \rightarrow X$. Let $i : X \rightarrow Y$ denote the inclusion. Then the induced map $i_* : H_k(X) \rightarrow H_k(Y)$ is an isomorphism for $k < n - 1$. In degrees $n - 1$ and n , we have the following exact sequence:*

$$0 \longrightarrow H_n(X, \mathbb{R}) \longrightarrow H_n(Y, \mathbb{R}) \longrightarrow H_{n-1}(\mathbb{S}^{n-1}, \mathbb{R}) \longrightarrow H_{n-1}(X, \mathbb{R}) \longrightarrow H_{n-1}(Y, \mathbb{R}) \longrightarrow 0.$$

Also note that $H_{n-1}(\mathbb{S}^{n-1}, \mathbb{R}) = H^{n-1}(\mathbb{S}^{n-1})^* = \mathbb{R}$ as computed in Example 5.6. In the case $n = 1$, assuming that both X, Y are connected and path-connected, we have

$$\begin{aligned} 0 \longrightarrow H_1(X, \mathbb{R}) \longrightarrow H_1(Y, \mathbb{R}) \longrightarrow H_0(\mathbb{S}^0, \mathbb{R}) = \mathbb{R}^2 \\ \longrightarrow H_0(X, \mathbb{R}) \oplus \mathbb{R} = \mathbb{R}^2 \longrightarrow H_0(Y, \mathbb{R}) = \mathbb{R} \longrightarrow 0. \end{aligned}$$

Proof. This is a result of the Mayer-Vietoris Theorem for homology (Theorem 5.8). The two open subsets $U, V \subset Y$ we use are

$$U = X \cup_g \left(e^n \setminus \overline{B\left(0, \frac{1}{3}\right)} \right), \quad V = B\left(0, \frac{2}{3}\right) \subset e^n \subset Y.$$

One can see that U can be contracted to X , and V is contractible (to a point). Also, their intersection is

$$U \cap V = \{x : r \in (1/3, 2/3)\} \subset e^n$$

which can be contracted to $\mathbb{S}^{n-1}$. The result then follows. $\square$

Example 6.3. Consider the n -dimensional complex projective space $\mathbb{CP}^n$, regarded as a $2n$ -dimensional real smooth manifold. We choose real numbers $c_0 < c_1 < \dots < c_n$, and define

$$f : \mathbb{CP}^n \rightarrow \mathbb{R}, \quad [z_0, z_1, \dots, z_n] \mapsto \frac{\sum_{j=0}^n c_j |z_j|^2}{\sum_{j=0}^n |z_j|^2}.$$

Consider the standard affine open coordinate chart $U_0 = \{[z_0, z_1, \dots, z_n] \in \mathbb{CP}^n : z_0 \neq 0\}$, with coordinates $w_1 = z_1/z_0 = x_1 + iy_1, \dots, w_n = z_n/z_0 = x_n + iy_n$. Then the function f on

U_0 in local coordinates is of the form

$$f|_{U_0} = c_0 \left(1 - \sum_{j=1}^n (x_j^2 + y_j^2) \right) + \sum_{j=1}^n c_j (x_j^2 + y_j^2) = c_0 + \sum_{j=1}^n (c_j - c_0)(x_j^2 + y_j^2).$$

Hence the Hessian of f under the coordinates $\{x_j, y_j\}_{j=1}^n$ is a diagonal matrix, with each of $c_j - c_0$ appearing twice as an eigenvalue. Since we assumed c_j to be distinct, f is non-degenerate on U_0 , with the only critical point $p_0 = [1, 0, \dots, 0]$. Since all the eigenvalues $c_j - c_0$ are positive, the index is $k_0 = 0$.

Similarly, any other affine open $U_j = \{z_j \neq 0\} \subset \mathbb{CP}^n$ contains only one critical point $p_j = [0, \dots, 0, 1, 0, \dots, 0]$, with the 1 at the j -th coordinate. The Hessian is diagonal with eigenvalues $c_i - c_j$, $0 \leq i \leq n$, $i \neq j$, each appearing twice. So there are in total $2j$ negative eigenvalues, and $\text{index}_f(p_j) = 2j$. Therefore $\mathbb{CP}^n$ has the same homotopy type with the CW complex $X = e^0 \cup e^2 \cup \dots \cup e^{2n}$.

We denote X_k the complex obtained after gluing the $2k$ -cell e^{2k} . By Lemma 6.2, we have an exact sequence

$$0 \longrightarrow H_{2k}(X_{k-1}, \mathbb{R}) \longrightarrow H_{2k}(X_k, \mathbb{R}) \longrightarrow \mathbb{R} \longrightarrow H_{2k-1}(X_{k-1}, \mathbb{R}) \longrightarrow H_{2k-1}(X_k, \mathbb{R}) \longrightarrow 0.$$

Since X_{k-1} only consists of cells of dimension at most $2k-2$, we have $H_{2k}(X_{k-1}, \mathbb{R}) = H_{2k-1}(X_{k-1}, \mathbb{R}) = 0$, so $H_{2k}(X_k, \mathbb{R}) = \mathbb{R}$ and $H_{2k-1}(X_k, \mathbb{R}) = 0$. By induction we also have $H_{2j}(X_k, \mathbb{R}) = \mathbb{R}$ and $H_{2j+1}(X_k, \mathbb{R}) = 0$ for $0 \leq j \leq k-1$. Hence

$$H^i(\mathbb{CP}^n) = \begin{cases} \mathbb{R} & \text{if } i = 2j, 0 \leq j \leq n, \\ 0 & \text{if } i = 2j-1, 1 \leq j \leq n. \end{cases}$$

Example 6.4. We compute the cohomology of the 2-torus $T = (\mathbb{S}^1)^2$, as an embedded submanifold in $\mathbb{R}^3$. The Morse function $f : T \rightarrow \mathbb{R}$ is the height function. More explicitly, we define

$$T = \{(x, y, z) = (a \cos \varphi + b \sin \theta \cos \varphi, a \sin \varphi + b \sin \theta \sin \varphi, b \cos \theta) : \theta, \varphi \in \mathbb{R}\}$$

$$f : T \rightarrow \mathbb{R}, (x, y, z) \mapsto x$$

where $a > b > 0$. Since the map $(\theta, \varphi) \mapsto (x, y, z)$ is an immersion, the critical points of f correspond to those with $\partial_\theta x = \partial_\varphi x = 0$. There are 4 critical points:

$$p_0 = (-a - b, 0, 0), \quad p_1 = (-a + b, 0, 0), \quad p_2 = (a - b, 0, 0), \quad p_3 = (a + b, 0, 0).$$

In particular, p_0 is the (local) minimum of f , p_1, p_2 are saddle points, and p_3 is the (local) maximum. So the respective indices are

$$\text{index}_f p_0 = 0, \quad \text{index}_f p_1 = \text{index}_f p_2 = 1, \quad \text{index}_f p_3 = 2.$$

Hence T is homotopy equivalent to the CW complex $X = e^0 \cup e^1 \cup e^1 \cup e^2$. If we denote by X_i the complex obtained after gluing the cell corresponding to $p_i, i = 0, 1, 2, 3$, then $X_1 \simeq \mathbb{S}^1$, so $H_0(X_1, \mathbb{R}) = H_1(X_1, \mathbb{R}) = \mathbb{R}$. By Lemma 6.2 the sequence is exact:

$$0 \longrightarrow \mathbb{R} \longrightarrow H_1(X_2, \mathbb{R}) \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \longrightarrow H_0(X_2, \mathbb{R}) = \mathbb{R} \longrightarrow 0.$$

So $H_1(X_2, \mathbb{R}) = \mathbb{R}^2$. The degree 2 homology of X_2 is trivial since it contains no 2-cells. By Lemma 6.2 again at degree 2, we have

$$0 \longrightarrow H_2(X, \mathbb{R}) \longrightarrow \mathbb{R} \longrightarrow \mathbb{R}^2 \longrightarrow H_1(X, \mathbb{R}) \longrightarrow 0.$$

On the other hand, since T is compact and orientable, by Poincaré duality we have $H_2(X, \mathbb{R}) \simeq H_2(T, \mathbb{R}) \simeq H^0(T) = \mathbb{R}$. Hence $H_1(X, \mathbb{R}) \simeq H_1(T, \mathbb{R}) \simeq H^1(T) = \mathbb{R}^2$, i.e.

$$H^\bullet(T) = \mathbb{R} \oplus \mathbb{R}^2 \oplus \mathbb{R}.$$

The result is consistent with the one obtained by applying Künneth's Formula, i.e. $H^\bullet(T) = H^\bullet(\mathbb{S}^1) \otimes H^\bullet(\mathbb{S}^1)$.

Example 6.5. As a generalization of Example 6.4, we compute the cohomology of the compact orientable surface Σ_g of genus $g \geq 0$, embedded in $\mathbb{R}^3$ with its “holes” aligned vertically. The Morse function $f : \Sigma_g \rightarrow \mathbb{R}$ we use here is again the height function. It has $2g + 2$ critical points $p_0, p_1, \dots, p_{2g+1}$ (corresponding to strictly increasing values of f), with minimum p_0 (of index 0), maximum p_{2g+1} (of index 2), and saddle points $p_1, \dots, p_{2g}$ (each of index 1). Hence Σ_g has the homotopy type of a CW complex X with one 0-cell, $2g$ 1-cells, and one 2-cell. Similarly as in the above example, we denote by X_k the complex obtained after gluing the cell corresponding to $p_k, k = 0, \dots, 2g + 1$. By Lemma 6.2 the sequence is exact for $1 \leq k \leq 2g$:

$$0 \longrightarrow H_1(X_{k-1}, \mathbb{R}) \longrightarrow H_1(X_k, \mathbb{R}) \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R}^2 \longrightarrow \mathbb{R} \longrightarrow 0.$$

By induction we have $H_1(X_k, \mathbb{R}) = \mathbb{R}^k$ for $0 \leq k \leq 2g$. For $X = X_{2g+1}$, at degree 2 we have

$$0 \longrightarrow H_2(X, \mathbb{R}) = \mathbb{R} \longrightarrow \mathbb{R} \longrightarrow \mathbb{R}^{2g} \longrightarrow H_1(X, \mathbb{R}) \longrightarrow 0$$

where $H_2(X, \mathbb{R}) = \mathbb{R}$ by Poincaré duality since Σ_g is compact and orientable. Hence $H_1(X, \mathbb{R}) = \mathbb{R}^{2g}$, and

$$H^\bullet(\Sigma_g) = \mathbb{R} \oplus \mathbb{R}^{2g} \oplus \mathbb{R}.$$

The result is consistent with the one computed using the fundamental group (recall that $\pi_1(\Sigma_g)$ is the quotient of the free group with $2g$ generators by one relation; the first homology group $H_1(\Sigma_g, \mathbb{Z})$ is its Abelianization, i.e. the free Abelian group $\mathbb{Z}^{2g}$ with $2g$ generators; and by the Universal Coefficient Theorem (Theorem 5.11) we have $H_1(\Sigma_g, \mathbb{R}) = H_1(\Sigma_g, \mathbb{Z}) \otimes \mathbb{R} = \mathbb{R}^{2g}$). Moreover its Euler characteristic is given by

$$\chi(\Sigma_g) = \sum_{i=0}^2 (-1)^i b_i = 1 - 2g + 1 = 2 - 2g.$$

References

- [1] R. Bott, *Morse Theory and its Application to Homotopy Theory* (LATEX-typed version, 2020), extracted from the following link: https://www.math.ucla.edu/~petersen/Bott_book.pdf
- [2] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology* (1982), Springer-Verlag New York, Inc.
- [3] A. Hatcher, *Algebraic Topology* (2002), Cambridge University Press.
- [4] J. Lee, *Introduction to Smooth Manifolds* (2013), 2nd edition, Springer.
- [5] J. Milnor, *Morse Theory* (1973), Princeton University Press.